



Demo Abstract: Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets

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Abstract

We present a privacy-preserving room occupancy estimation method using federated analytics of Bluetooth Low Energy (BLE) packets. By processing data locally and reporting only aggregated device counts, our approach preserves user privacy while achieving 95% accuracy in occupancy prediction. This method provides a cost-effective alternative to traditional sensing technologies like PIR and mmWave, balancing privacy, accuracy, and ease of deployment. Future work will expand testing to multi-room setups and enhance privacy measures. Code URL: <https://github.com/Johnnybyzhang/BLE-capture>

CCS Concepts

• **Computer systems organization** → **Sensors and actuators; Embedded systems; Sensor networks**; • **Security and privacy** → **Privacy protections**.

Keywords

Room occupancy, BLE, federated analytics, privacy-preserving, sensor networks, packet filtering

ACM Reference Format:

Boyan Zhang, Chenshuhao Qin, and Bing Luo. 2024. Demo Abstract: Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets. In *ACM Conference on Embedded Networked Sensor Systems (SenSys '24)*, November 4–7, 2024, Hangzhou, China. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3666025.3699422>

1 Introduction

1.1 BLE and Occupancy Estimation

Bluetooth Low Energy (BLE) is a wireless communication standard designed for resource-constrained devices. It has been integrated into numerous everyday devices, including smartphones, tablets, headphones, and health trackers, among others. In 2022 alone, over 3 billion BLE-enabled devices were shipped, demonstrating its widespread adoption and importance in various IoT applications. [7]

One emerging application of BLE is in occupancy detection systems. Traditional methods often rely on costly hardware and raise

This work is supported by Kunshan Municipal Government research funding (Project 24KKSGR013). Corresponding Author: Bing Luo.

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SenSys '24, November 4–7, 2024, Hangzhou, China
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ACM ISBN 979-8-4007-0697-4/24/11
<https://doi.org/10.1145/3666025.3699422>

privacy concerns. In contrast, BLE offers a low-cost, efficient alternative by detecting variations in Bluetooth signals related to human presence. Research shows that this approach can classify occupancy with an accuracy of 97.97% and estimate the number of people in a room with an average difference of just 0.32 persons from the actual count, making it a practical solution for energy-saving automation in public buildings [1].

1.2 Privacy Concerns

The use of BLE packets for occupancy estimation is not without risks. Previous studies have shown that BLE transmitters have unique Carrier Frequency Offsets (CFOs) that can be easily fingerprinted by passive adversaries, potentially compromising user location privacy [6].

1.3 Existing Solutions and Limitations

Current approaches to sensor-based occupancy estimation vary in their effectiveness, privacy protection, and practicality. The table below summarizes some existing methods and their limitations:

Method	Installation	Privacy	Static User	Speed	Accuracy
PIR[5]	Ceiling	High	false positive	Fast	High
mmWave[4]	Line of Sight	High	high sensitivity	Fas	High
Wi-Fi	Anywhere	Poor	Normal	Slow	Lowest
BLE[1]	Anywhere	Poor	Normal	Moderate	Moderate
BLE (FA)	Anywhere	High	Normal	Moderate	Moderate

Table 1: Comparison of different sensing methods.

2 Proposed Method

We propose a privacy-preserving approach to room occupancy estimation using federated analytics of BLE packets. Our method addresses the limitations of existing solutions while maintaining high accuracy and ease of deployment.

2.1 System Architecture

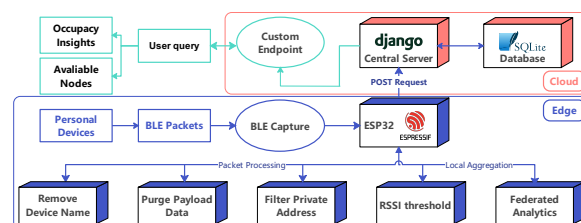


Figure 1: Demo System Diagram

Our system employs a single BLE packet capture node that processes data locally and reports only aggregated information, adhering to federated analytics principles. Key features include:

- **MAC Address Whitelist:** Filters out "Virtual Addresses" to reduce invalid devices.

- Payload Protection: Packet payloads are never decoded or transmitted.
- Aggregated Reporting: Only the count of unique devices is reported, preserving individual privacy.

2.2 Demonstration Setup

We conducted a proof-of-concept demonstration in a real-world setting for a duration of 1.5 hours, along with additional telemetry data to better illustrate the effectiveness of the setup:

- Equipment: ESP32-S3 Development Board
- Data Collection: Continuous capture of BLE packets, filtered and processed in real-time
- Reporting: aggregated device counts and total packet count every 30 second

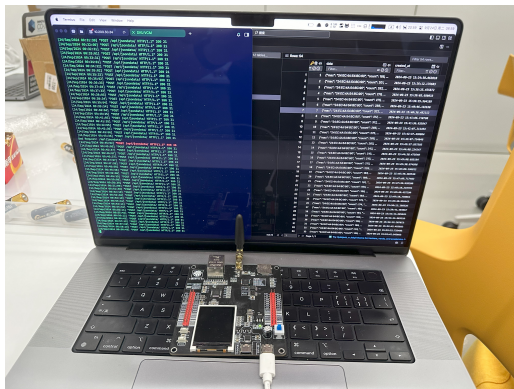


Figure 2: Image of the Demo Setup

3 Results and Discussion

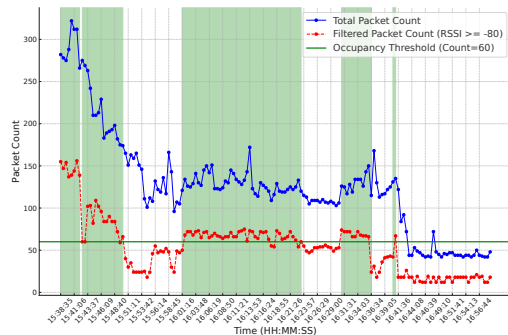


Figure 3: Room occupancy result via BLE packets

3.1 Result Overview

Our demonstration of the BLE-based federated analytics approach for room occupancy estimation yielded compelling results. The graph illustrates the relationship between BLE packet counts and room occupancy over time. We observed:

- A clear distinction between total packet count (blue line) and filtered packet count (red dashed line, $RSSI \geq -80$).
- Green highlighted sections indicating periods when the room was known to be empty based on ground truth data.

Using a threshold of 60 on the filtered BLE packet count, our model achieved a remarkable 95% accuracy in predicting room occupancy.

This high accuracy demonstrates the potential of our approach for reliable occupancy sensing. The results break down as follows:

- True Positives: 80 instances True Negatives: 72 instances
- False Positives: 7 instances False Negative: 1 instance

3.2 Privacy Protection

Our federated analytics approach successfully maintained user anonymity by only reporting aggregated device counts, avoiding the privacy concerns associated with conventional BLE tracking methods. The central server has no information on specific device locations, yet with the power of federated analytics, we can gain insight into the status of the room. Besides making use of federated analytics, our project does not allow any captured packet outside of RAM on the ESP32, further making sure that none of the sensitive information could be extracted even if the central server got compromised.

4 Conclusion and Future Work

This demo showcases a novel approach to room occupancy estimation using federated analytics of BLE packets. Our method offers a balance between privacy protection and accurate occupancy insights, addressing the limitations of existing technologies. Future work will focus on:

- Extended testing duration and multi-room setups
- Exploration of advanced privacy-enhancing techniques within the federated analytics framework
- Adding an on-device algorithm that actively tweaks the threshold of packets based on local information
- Integrate with the existing FedCampus setup to make full use of data and compute from endpoint client devices [2]
- Employ a layered network so more relevant data can be accessed on lower-level SBC servers that can also do Federated Learning based on its connected nodes [3]

Our demonstration lays the groundwork for privacy-conscious occupancy sensing systems that can be widely deployed without compromising user privacy.

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